

	power dissipated across X and Y increased. Hence ratio increased.
4(a)	By Fleming's left hand rule, the moving charge experiences a magnetic force which is always perpendicular to its velocity and the magnetic flux density. This force provides for the charge's centripetal acceleration and changes the charge's direction of travel but not its speed. With the speed remaining constant, the magnetic force always point towards the centre of the charge's circular path.
4(b)(i)	The electric field produces an electric force which cancels the magnetic force. Hence the charge moves at constant velocity. $F_E = F_B$ $qE = Bqv$ $E = Bv$ $= (3.2 \times 10^{-3})(4.7 \times 10^5)$ $= 1.5 \times 10^3 \text{ V m}^{-1}$
4(b)(ii)	Magnetic force is directly proportional to velocity but electric force does not depend on velocity.